APEDP: Scientific Architecture, Assessment & Action Plan

System: APEDP (Autonomous Physics & Epistemic Discovery Platform) **Assessment Date:** September 30, 2026 **Status:** Finalized Consensus Document

Maturity Rating: 8.8 / 10 (Research System / Prototype)

Judged strictly on observed code behaviors, mathematical properties, epistemic contracts, and test executions in live sessions.

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Core Axiom: Discovery ≠ Fitting ≠ Selection ≠ Adjudication
APEDP Architecture = Discovery Engine + Evidence Engine + Epistemic Adjudicator + Intervention Layer
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1. Dimensional Evaluation Scorecard

Dimension	Rating	Observed Evidence & Rationale
Epistemic Architecture	9.4 / 10	Robust 4-state adjudication (UNIQUE, MULTI_HYPOTHESIS, INSUFFICIENT, NO_VALID_CANDIDATE); strictly separates numerical best-fit from explanatory evidence.
False-Promotion Resistance	9.1 / 10	Guard 1 absolute fidelity ceiling (3.0·σ/CAL) blocks before comparison; Guard 2 (|ρ1|≤0.5) residual whiteness barrier prevents overclaiming.
Validation Discipline	9.2 / 10	Strict train/holdout separation; paired block bootstrap flips ambiguous point-margins to MULTI when CI straddles 0; real SE series tracking.
Scientific Constraints	9.0 / 10	Dimensional homogeneity gates; Lie symmetries; SI unit consistency; physically motivated structural candidate representations.
Discovery Architecture	8.3 / 10	Proposer abstraction and grammar discovery exist and recover Duffing/coupled ODEs, but broader unconstrained open discovery requires scaling.
Real-World Evidence	8.7 / 10	Bertozzi (1964) Linac electron calorimeter experimental validation; FRED macroeconomic observational series; Trapped-ion quantum simulation.
Active / Intervention	8.2 / 10	Closed-loop perturbation demonstrated; Fisher Information Matrix (FIM) experiment design exists but L3 active discrimination needs maturation.
Software Engineering	9.1 / 10	High performance core; 407+ automated tests (38/38 core regressions, 34/34 constitution, 34/34 sandbox/realdata); zero static lint defects.
Benchmark Maturity	8.8 / 10	Multi-domain suites (materials, quantum dynamics, market time series, plasmas); eliminated mock losses; requires broader external suites.

Dimension	Rating	Observed Evidence & Rationale
Reproducibility & Governance	9.3 / 10	Immutable frozen hashes; version stamps (bootstrap-v3+solefid-v5); CAPABILITY_ABSENT probes; 100% equivalence checks on self-improvement gates.

Key Epistemic Principle Formally Established:

argmin_m RMSE(m) ≠ argmax_m Evidence(m) &implies; point-estimate winner ≠ UNIQUE_SUPPORT

APEDP demonstrates that having the lowest observed holdout error does not grant a model promotional rights when observational uncertainty or collinearity cannot rule out competing explanations.

2. Finalized Architectural Decisions

Decision 1: Standardized 4-Tier Provenance Taxonomy (`DataOrigin`)

Formally codified in `apedp_dyn/epistemic/adjudicate.py` to ensure scientific transparency across all data sources:

- `SYNTHETIC`: Analytical mathematical evaluations with synthetically injected noise (e.g., Duffing oscillator benchmark).
- `SIMULATION`: Numerical ODE/PDE forward integrations or analog quantum Hamiltonian simulations (e.g., 13-ion trapped-ion string-breaking simulator).
- `REAL_OBSERVATIONAL`: Passive observational field or market records collected without controlled interventions (e.g., FRED macroeconomic time series, S&P 500).
- `EXPERIMENTAL`: Active laboratory or clinical interventions with manipulated independent variables (e.g., Bertozzi 1964 Linac electron calorimeter).

Decision 2: Formalized 4-Level Benchmark Capability Hierarchy (`BenchmarkLevel`)

Benchmarks must explicitly declare the capability tier under test to prevent fixed-model selection from masquerading as equation discovery:

Capability Level	Candidate Source	Capability Under Evaluation
L0 — Selection	Predefined / Human candidate sets	**Anti-Overclaiming:** Can APEDP compare competing models without overclaiming when evidence is ambiguous?
L1 — Constrained Discovery	Restricted domain grammar	**Term Isolation:** Can APEDP generate plausible structures from vocabulary and isolate active physical terms?
L2 — Open Discovery	Broad grammar + physical gates	**Generative Recovery:** Can APEDP discover previously unspecified ODEs/PDEs across Pareto complexity frontiers?
L3 — Active Discovery	L2 + closed-loop intervention	**Intervention Design:** Can APEDP design the optimal measurement that actively discriminates remaining hypotheses?

Decision 3: Canonical S&P 500 Benchmark Preserved as L0 Real-Observational

The financial time series sandbox (`apedp_dyn/sandbox/finance_demo.py`) is frozen as a canonical L0 benchmark:

- **COVID Rebound:** Successfully identified momentum AR(1) dynamics ($P_t = 268.1 + 0.9096\ P_{t-1}$) over static trend and constant models.
- **2022 Market Regimes:** AR(1) had lowest point RMSE, but APEDP correctly returned `MULTI_HYPOTHESIS` because epistemic margins did not clear the resolution floor τ_{epi}. This abstention is preserved as a core capability proof.

Decision 4: Built & Verified L1 Constrained Materials Discovery Benchmark

Implemented in `apedp_dyn/sandbox/materials_discovery_l1.py` and verified in `tests/test_dyn_materials_l1.py`:

- **M1 (Single Mechanism):** $\sigma_y = \sigma_0 + k \cdot d^{-1/2}$. Promoted `hp_single` with `UNIQUE_SUPPORT` (Δ=70.8 > τ=9.76).
- **M2 (Competing Mechanisms):** $\sigma_y = \sigma_0 + k \cdot d^{-1/2} + A \cdot \Phi^{1/2}$. Promoted `dual_hp_precip` with `UNIQUE_SUPPORT` (Δ=25.1 > τ=13.8).
- **M3 (Mechanism Overlap / Collinearity):** Microstructural Zener pinning couples $\Phi \sim 1/d$, making $d^{-1/2}$ and $\Phi^{1/2}$ collinear. APEDP properly abstained with `MULTI_HYPOTHESIS` (Δ=2.31 ≤ τ=11.3), refusing to award UNIQUE support to a point-estimate winner.

Decision 5: Trapped-Ion Quantum Matter Sandbox

Implemented in `apedp_dyn/sandbox/quantum_vacuum_sandbox.py` based on the Duke Quantum Center 13-ion lattice gauge simulation (Nature Physics 2026), testing non-perturbative Schwinger tunneling vs power-law models and spatiotemporal string breaking.

3. Immediate Action Plan & Elevation Roadmap (Path to 9.2 – 9.5+)

To bridge the remaining ~1.2 points and advance APEDP from an 8.8 research prototype to a 9.2–9.5+ mature scientific discovery engine, four targeted engineering and research initiatives are established:

Action Item	Core Deliverable	Target Milestone & Acceptance Gate
1. Open Discovery Expansion (L1 → L2)	Extend grammar enumeration from restricted vocabulary to multi-variable Lie-bracket & dimensional invariants.	Unconstrained equation discovery recovering nonlinear coupled PDEs/ODEs without pre-supplied equation templates.
2. Calibrated Uncertainty Quantification	Replace uncalibrated parameter SE estimators with profile-likelihood contour profiling and block bootstrap parameter distribution propagation.	Empirical coverage tests demonstrating parameter and trajectory 95% confidence intervals achieve ≥93% empirical coverage across noise grids.
3. Independent Blind External Validation	Establish a pre-registered benchmark runner executing outside the internal development harness on withheld empirical datasets.	Pre-registered evaluation against 20+ blinded physical datasets (NIST / external experimental records) with zero developer-in-the-loop tuning.
4. Active Closed-Loop Intervention (L3)	Couple Fisher Information Matrix (FIM) optimal experimental design directly to MULTI_HYPOTHESIS states.	When APEDP abstains with MULTI_HYPOTHESIS, it automatically synthesizes the minimal discriminatory intervention (e.g. specific d or T perturbation) to resolve the tie.

4. Live Test & Verification Summary

All core deliverables, provenance updates, benchmark levels, and new discovery suites are verified green:

- **Core Adjudication Regressions:** 38/38 passed (17.07s) in `tests/test_dyn_adjudicate_regression.py`.
- **Constitution & Invariant Gates:** 34/34 passed in `tests/test_constitution.py`; 1027/1027 equivalence agree.
- **L1 Materials Discovery Suite:** 6/6 passed (1.24s) in `tests/test_dyn_materials_l1.py`.
- **Quantum Vacuum Sandbox Suite:** 11/11 passed in `tests/test_dyn_quantum_sandbox.py`.
- **Central Sandbox & Real Data Suite:** 26/26 passed (1.50s) in `tests/test_dyn_sandbox.py`.
- **Static Code Hygiene:** `ruff check` reports **0 errors** across all modified modules.

APEDP Autonomous Physics & Epistemic Discovery Platform • Session Frozen Baseline: `bootstrap-v3+solefid-v5` • Generated: 2026-09-30